

SUMMARY TABLE: GENERAL SCIENCE GRADE 10

Sub-Strand 2.5: Rates of Reactions — Teacher Reference (Pre-filled)

SUMMARY TABLE: GENERAL SCIENCE GRADE 10	
Sub-Strand	Sub-Strand 2.5: Rates of Reactions
Driving Question	DRIVING QUESTION: Why do some chemical reactions happen in a flash while others take days — and how can we control the speed of reactions to make life in Kenya safer, healthier, and more productive?

INSTRUCTIONS
FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 What Is a Rate of Reaction? Launching the Disappearing Cross	In the demo, the cross vanished at 35 s in warm Beaker B (50 C) and at 270 s in cold Beaker A (10 C). A yellow precipitate of sulfur formed in both. The chemicals and amounts were the same and only temperature differed, so B was 7.7 times faster.	Rate of reaction is how fast reactants are used up or products are formed. It can be measured by the time for a change (such as a precipitate hiding a cross), by the volume of gas produced, or by loss of mass.	So far we can only say that temperature seems linked to speed and that particles may be involved. We have a first particle sketch and DQB questions, but we cannot yet explain why the warm reaction was faster.
Lesson 2 Measuring Reaction Rates: Designing a Fair Test	We calculated the rate for the demo: Beaker B (50 C, 35 s) has rate $1/35 = 28.6 \times 10^{-3} \text{ s}^{-1}$ and Beaker A (10 C, 270 s) has rate $1/270 = 3.7 \times 10^{-3} \text{ s}^{-1}$, so B is about 7.7 times faster. Different observers judged the cross gone at slightly different times.	Rate = $1/t$ for the cross method, so a shorter time gives a bigger rate. In a fair test we change only the independent variable (temperature), measure the dependent variable (time, then rate) and keep control variables the same (50 cm ³ thiosulfate, 10 cm ³ HCl, beaker, cross, observer).	We can compare the two beakers fairly because the chemicals, amounts and way of judging the cross were the same and only temperature differed. We cannot yet explain WHY the warm beaker reacts faster; that question stays on the DQB for Lesson 3.
Lesson 3 Factor 1: Temperature	What did we observe? Record the class-mean times for the cross to disappear at 10 C (270 s), 25 C (110 s), 40 C (52 s) and 55 C (26 s), and the rates 3.7, 9.1, 19.2 and $38.5 \times 10^{-3} \text{ s}^{-1}$. Note how the demo at 50 C (35 s) fits, and that the amount of sulfur made was the same.	What did we learn? Write what happens to the rate of the thiosulfate-HCl reaction as temperature rises, and how the rate changes from 10 C to 55 C. State that temperature changes the speed of the reaction and not the final amount of product.	How can we explain it? Use particles to explain why a higher temperature increases the rate: more kinetic energy, more frequent and more energetic collisions, a larger fraction of effective collisions (activation energy). Use this to explain why warm Beaker B was faster than cold Beaker A.
Lesson 4	We observed that at 25 C the cross	We learned that a higher concentration	We explained that in a more concentrated

Factor 2: Concentration	disappeared in 110 s, 138 s, 183 s and 275 s as the sodium thiosulfate concentration decreased from 0.15 to 0.12, 0.09 and 0.06 mol/dm ³ . The matching rates (1/t) were 9.1, 7.2, 5.5 and 3.6 x10 ⁻³ s ⁻¹ . The weaker the solution, the longer the cross stayed visible.	means more particles per unit volume, and that rate is roughly proportional to concentration (rate / concentration is about 60 for every row). Concentration changes how fast the reaction goes, not the type of products, and it is a second factor besides temperature.	solution there are more particles in the same volume, so they collide more often, giving more effective collisions per second and a faster rate. The energy of the particles did not change because the temperature stayed at 25 C. In the anchor demo the concentration was the same in Beakers A and B, so temperature, not concentration, caused the 35 s versus 270 s difference.
Lesson 5 Factor 3: Surface Area – Why Does Powder React Faster Than Chunks?	Powder fizzed fastest and levelled off at 120 cm ³ of CO ₂ ; small chips were next; large chips were slowest. Initial rates over the first 30 s were 0.40 cm ³ /s (large), 0.93 cm ³ /s (small) and 2.07 cm ³ /s (powder).	Breaking a solid into smaller pieces increases its surface area, so more particles can collide with acid particles and the rate rises. The gradient of a volume-time curve gives the rate. Surface area changes the speed, not the final amount of gas (maximum about 120 cm ³).	With the same mass, acid and temperature, powder exposes more CaCO ₃ particles to HCl particles, so there are more effective collisions per second and a steeper curve. This is the same collision idea used for temperature and concentration, and it helps explain why unga cooks faster than whole maize and why a crushed antacid works quickly.
Lesson 6 Factor 4: Catalysts and Light	In 30 s, 10 cm ³ of 3% hydrogen peroxide with a few drops of detergent produced foam of 0 cm with no additive, 0 cm with sand, 0.5 cm with a boiled liver cube and 9 cm with a fresh liver cube (1 cm ³ each). The fresh liver cube was still there afterwards and worked again in new peroxide. Temperature, concentration and amount were the same in every cylinder.	A catalyst speeds up a reaction without being used up and does not change the final amount of product. The enzyme catalase in fresh liver is a biological catalyst; boiling damages it. Light is an energy source for some reactions (photosynthesis) and is not a catalyst. Catalysts are used in industry, such as the iron catalyst in the Haber process for ammonia.	A catalyst gives the reaction an alternative pathway with a lower activation energy, so a larger fraction of collisions have enough energy to be effective collisions, and the rate rises without changing temperature, concentration or size. This adds to the Disappearing Cross story: warm Beaker B (35 s) beat cold Beaker A (270 s) because heating gives particles more energy, while a catalyst lowers the energy needed.
Lesson 7 Applying the Model to Biological, Chemical and Physical Processes in Kenya	We looked at the Beaker A / Beaker B result (10 C = 270 s; 50 C = 35 s) and compared it with Kenyan examples: uji fermenting in about 24 h at 30 C but about 48 h at 15 C, a fridge at 4 C, amylase working best near 37 C and pH 7, pepsin near pH 2, dried and salted fish, and rusting roofing sheets in Mombasa.	Several factors act together in real systems. Enzymes (biological catalysts) have optimum temperature and pH. Cooling, drying and salting slow the reactions that spoil food. The factors change the speed of a reaction, not the final amount made.	Because cooler conditions give particles less kinetic energy, fewer collisions have enough energy (activation energy) to react, so there are fewer effective collisions per second and the rate is lower. Enzymes work best when their shape suits the temperature and pH. Drying and salting remove the water needed for spoilage reactions, so food keeps longer.
Lesson 8 Model Building, Assessment and Community Connection: The Final Explanation	In the anchor demonstration, the cross vanished at 35 s in warm Beaker B (50 C) and at 270 s in cold Beaker A (10 C): 7.7 times faster, with rates 28.6 and 3.7 x10 ⁻³ s ⁻¹ . The same chemicals and amounts were used, and both ended with the same	Reaction rate depends on how many effective collisions happen per second. Higher temperature, higher concentration, larger surface area and a catalyst all increase the rate; light supplies energy for some reactions. Factors change how fast,	Write your Final Explanation: Beaker B was faster because at 50 C the particles have more kinetic energy, so they collide more often and with more energy, and a larger fraction of collisions have at least the activation energy, making them effective.

	amount of sulfur. Add one row of evidence from the temperature, concentration or surface area table.	not how much, product forms. Optimum conditions exist in biological systems, and cooling, drying and salting slow reactions to preserve food.	Cite your evidence row. Then explain how you would change one factor to solve a real Kenyan problem (for example preserving fish at Lake Victoria) and how your model has changed since Lesson 1.
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ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

Production & Layout: Educational design and curriculum sequencing by SeaVuria. Technical layout, structural formatting, and digital optimization/implementation executed in partnership with ARES Education (registered 501(c)(3) in the United States and Canada (as Afretech Aid) and registered NGO in Kenya), with lesson drafting and compilation assisted by AI (Claude, Anthropic).

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